

Image Math

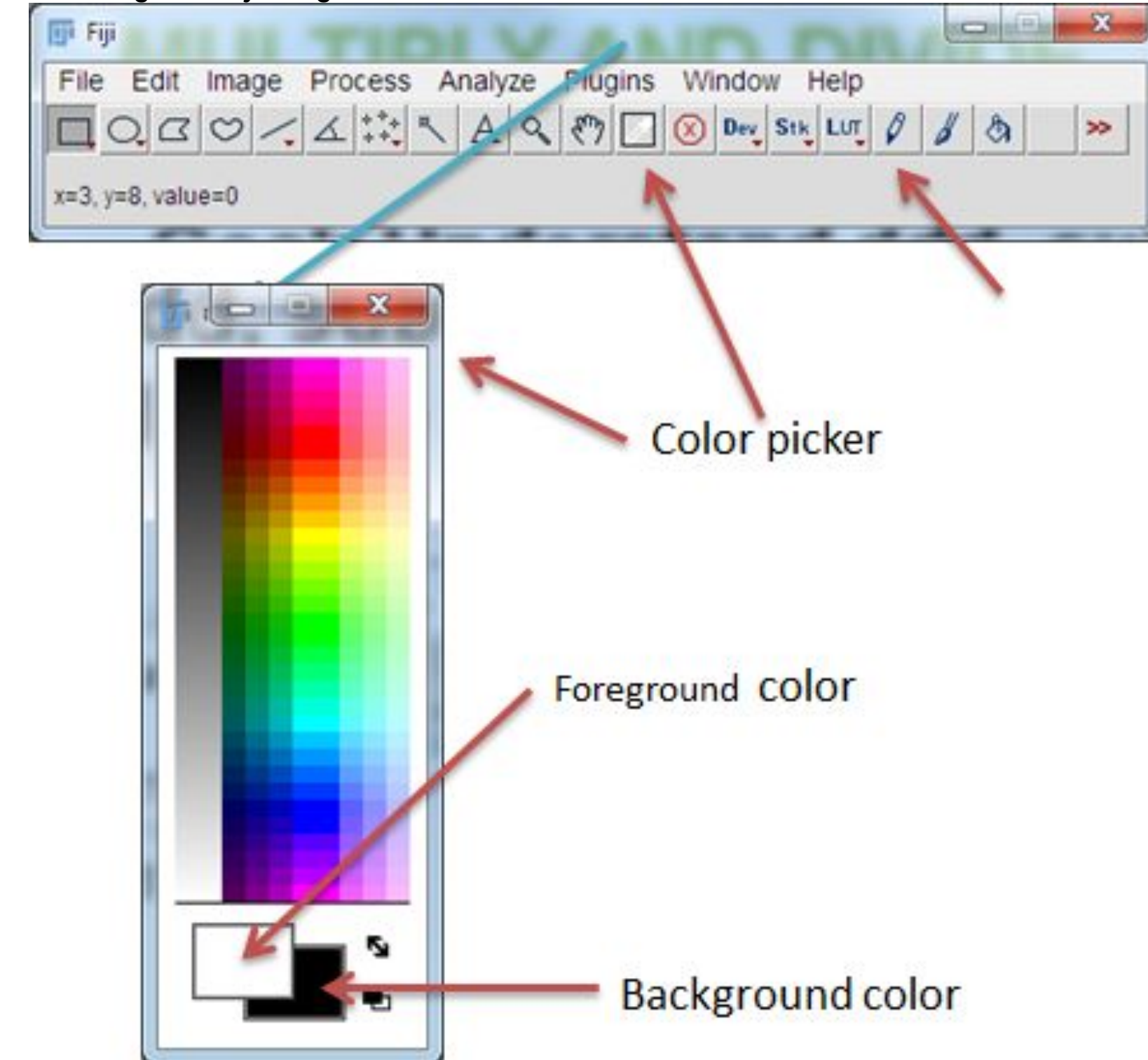
Image Processing & Analysis for Life Scientists

Olivier Burri, Romain Guiet & Arne Seitz

Global Operations

1. Create a new 8-bit image and draw on it with multiple values
Put the pointer on the image and look at the values

Drawing with Fiji/ImageJ

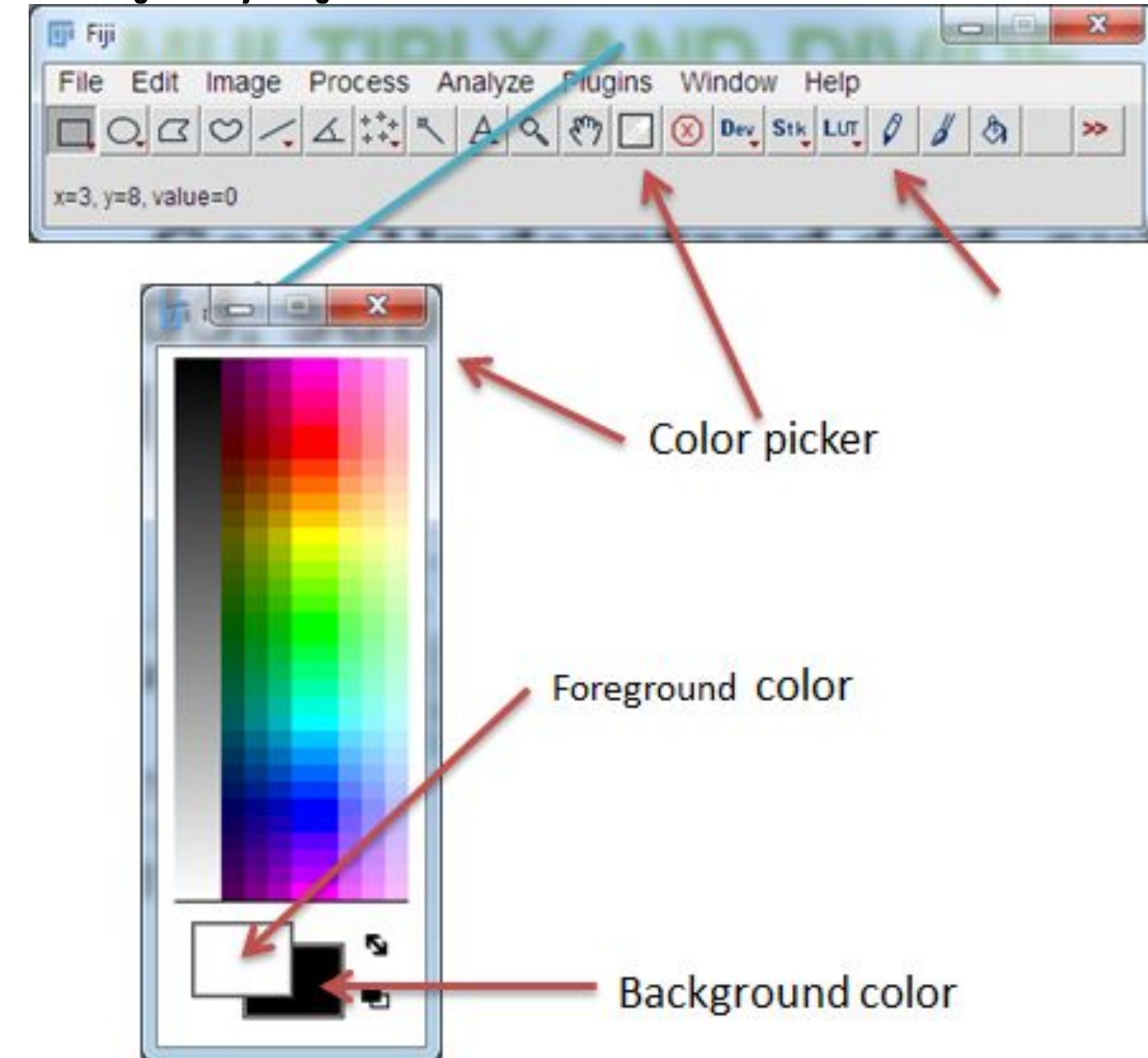


NOTE: Although the color picker offers you colors, images in 8-bit, 16-bit and 32-bit depth are grayscale images and color will not show up on them. The only way to see grayscale in color is by using lookup tables

Global Operations

1. Create a new 8-bit image and draw on it with multiple values
Put the pointer on the image and look at the values
2. **Add 10** to the image: **Process>Math>Add**
3. Look at the pixels values with the cursor. What happens to values already at 255?

Drawing with Fiji/ImageJ

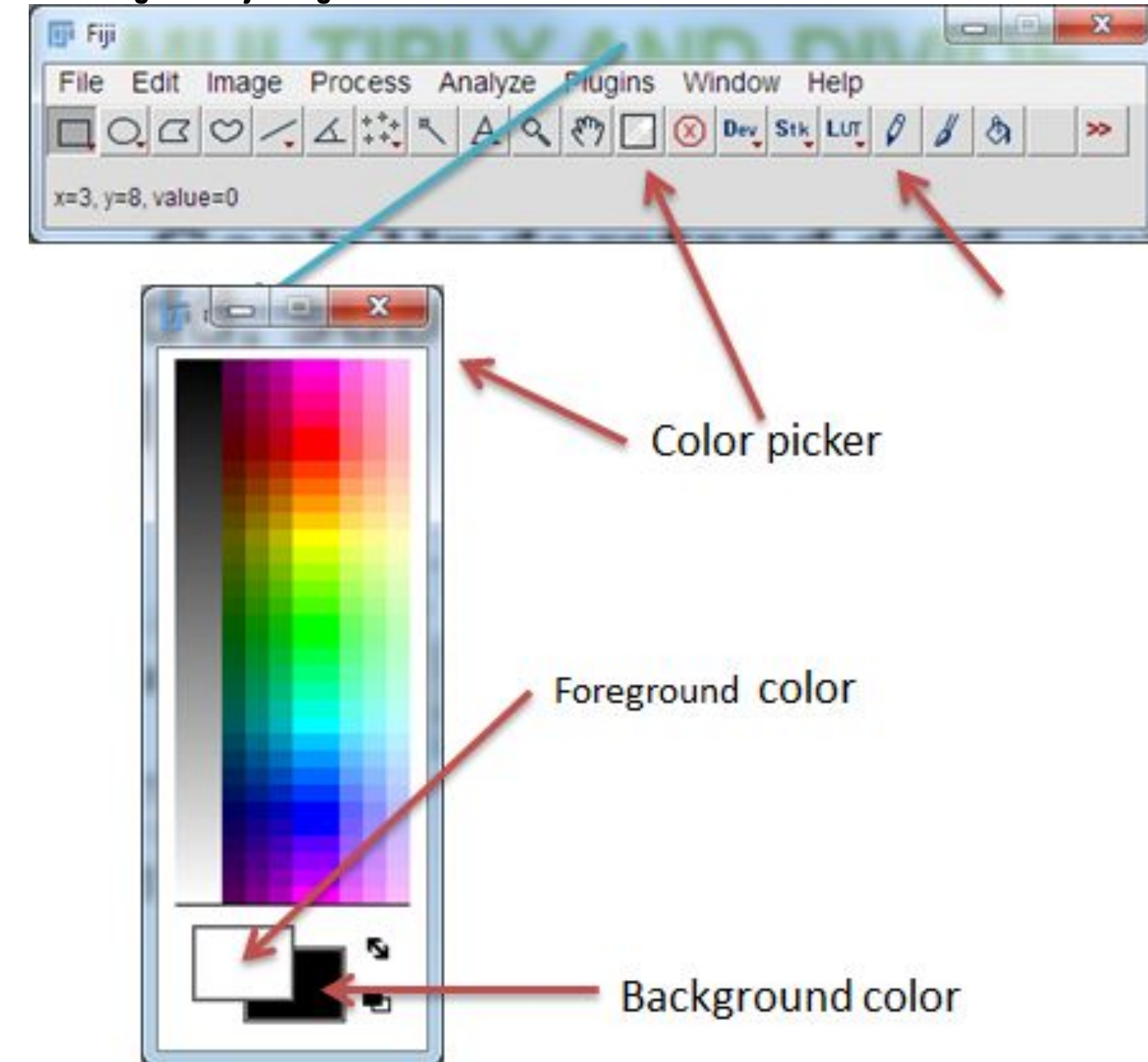


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Put the pointer on the image and look at the values
2. **Add 10** to the image: **Process>Math>Add**
3. Look at the pixels values with the cursor. What happens to values already at 255?
4. **Create or open** another 8-bit image
Add 100 Process>Math>Add
Divide by 3 Process>Math>Divide
5. Notice the rounding of pixel values

Drawing with Fiji/ImageJ

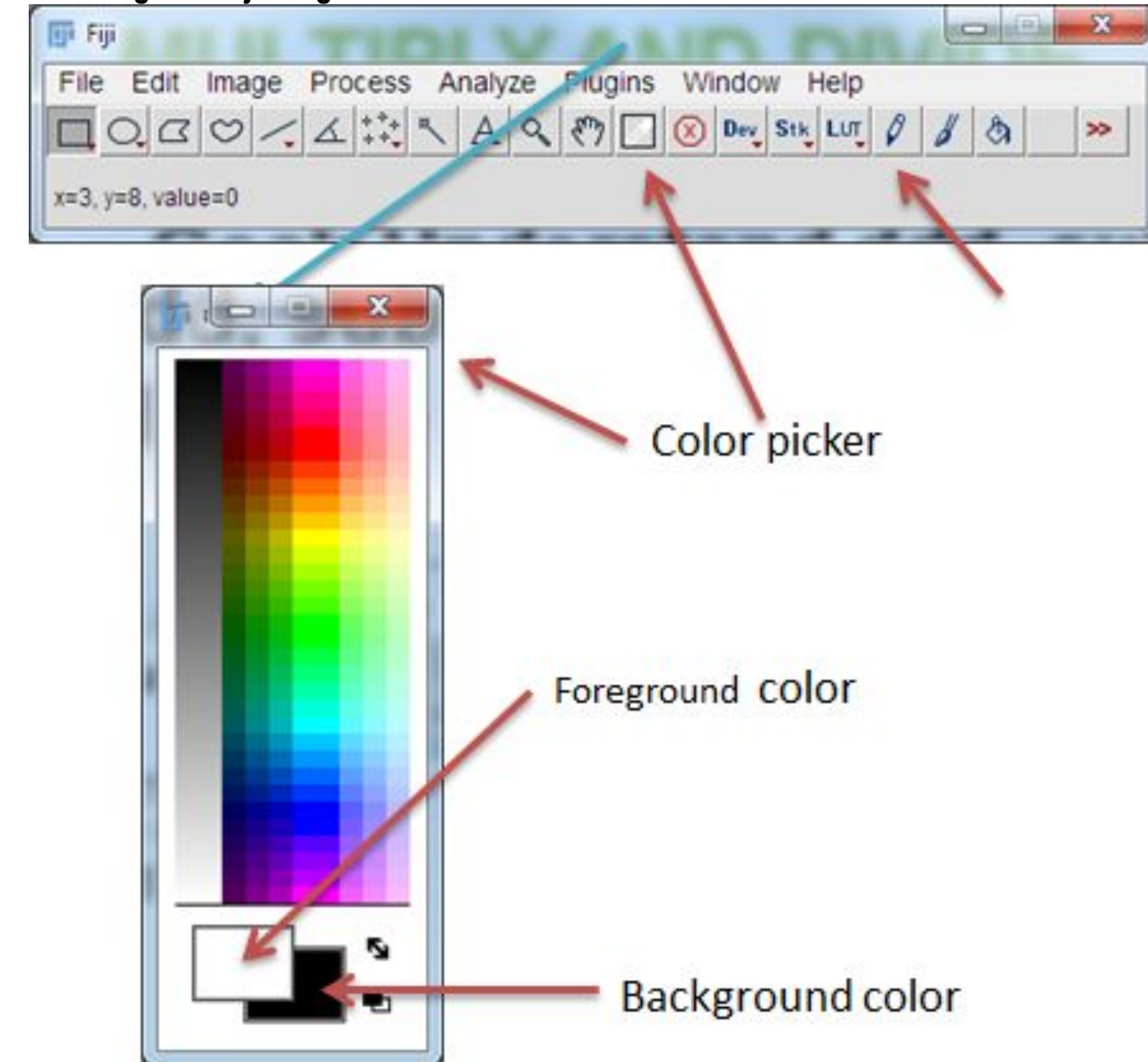


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6. Try 4. again on a **32-bit image**

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Image to Image Operations

1. [Download the dataset](#) and open both images
 - 63x pinhole 1AU corner zoom0.5 3channels.tif
 - Normalized Flatfield 63x pinhole 1AU corner zoom0.5 3channels.tif
2. Note how the intensities drop at the edges of the image for all channels.
3. Look at the **Normalized Flatfield** image.

This image was acquired by averaging many image fields together to get the global unevenness of the field of view. The intensities are normalized to the maximum value of each channel.
- 4.

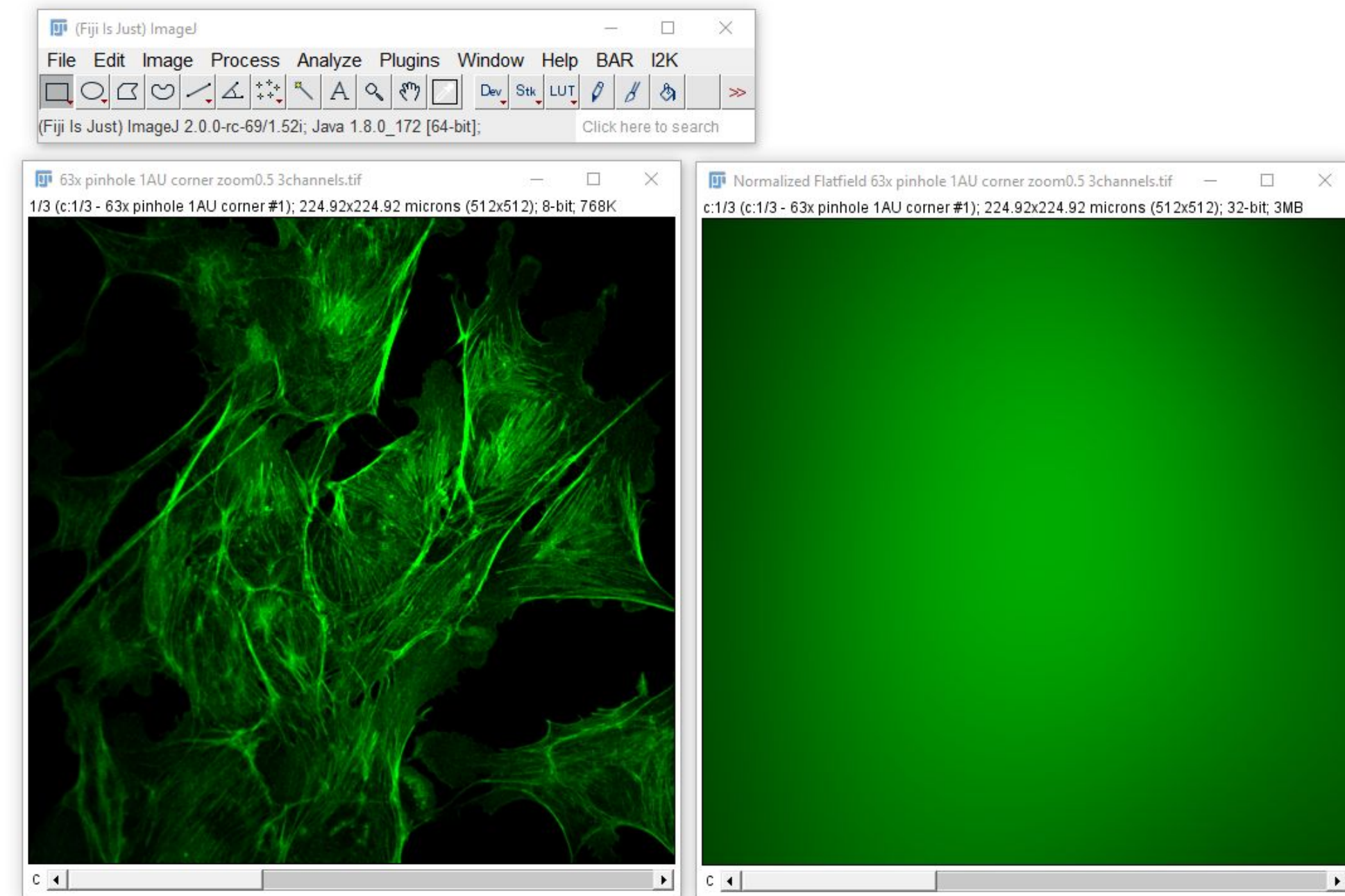


Image to Image Operations

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3. Look at the **Normalized Flatfield** image.
This image was acquired by averaging many image fields together to get the global unevenness of the field of view. The intensities are normalized to the maximum value of each channel
4. Perform an image division
Process>Image Calculator

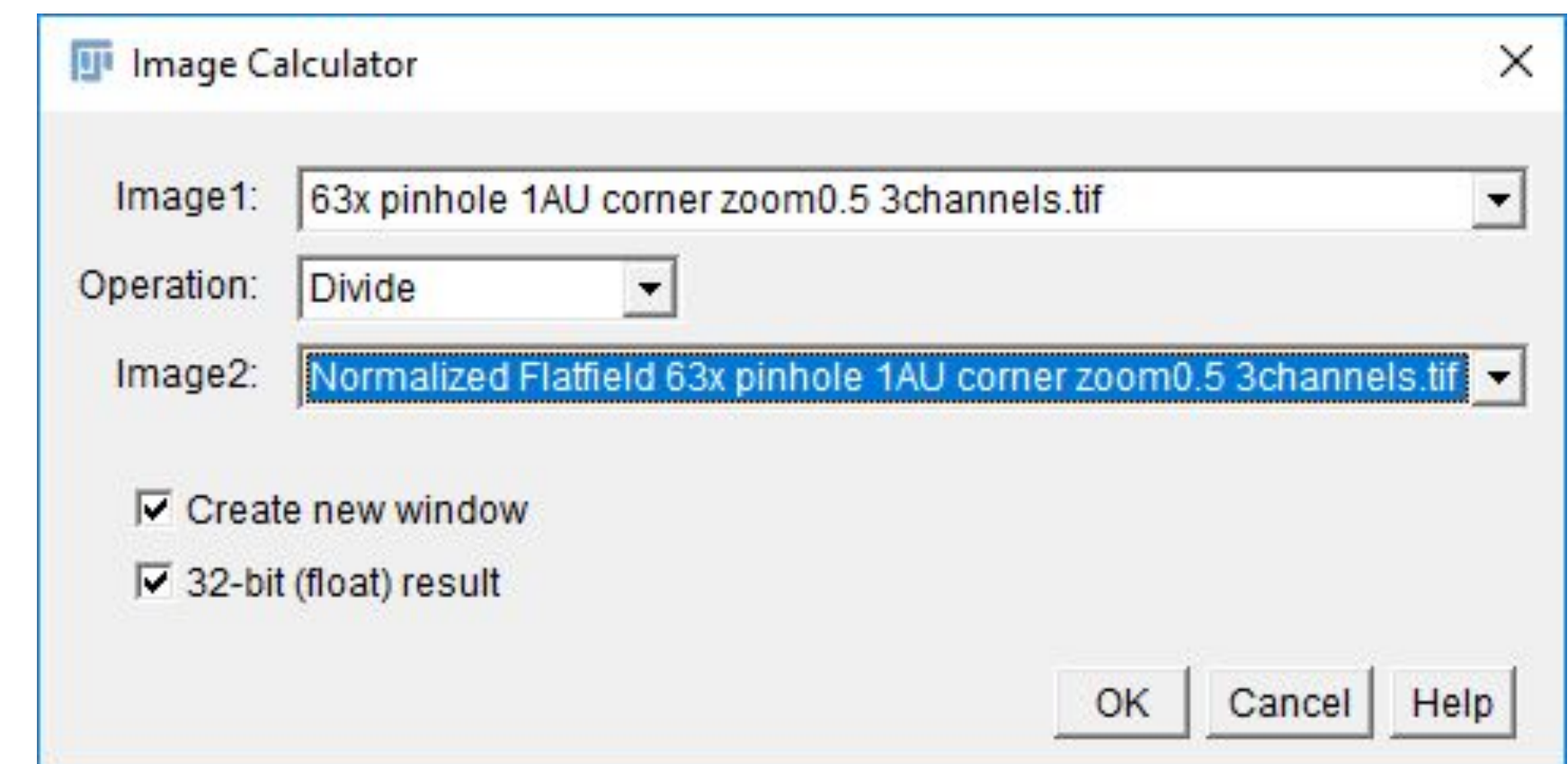
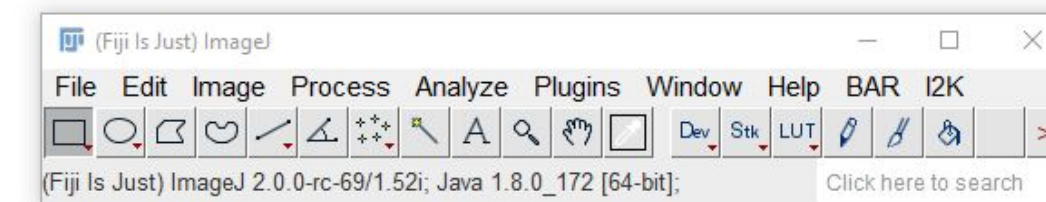
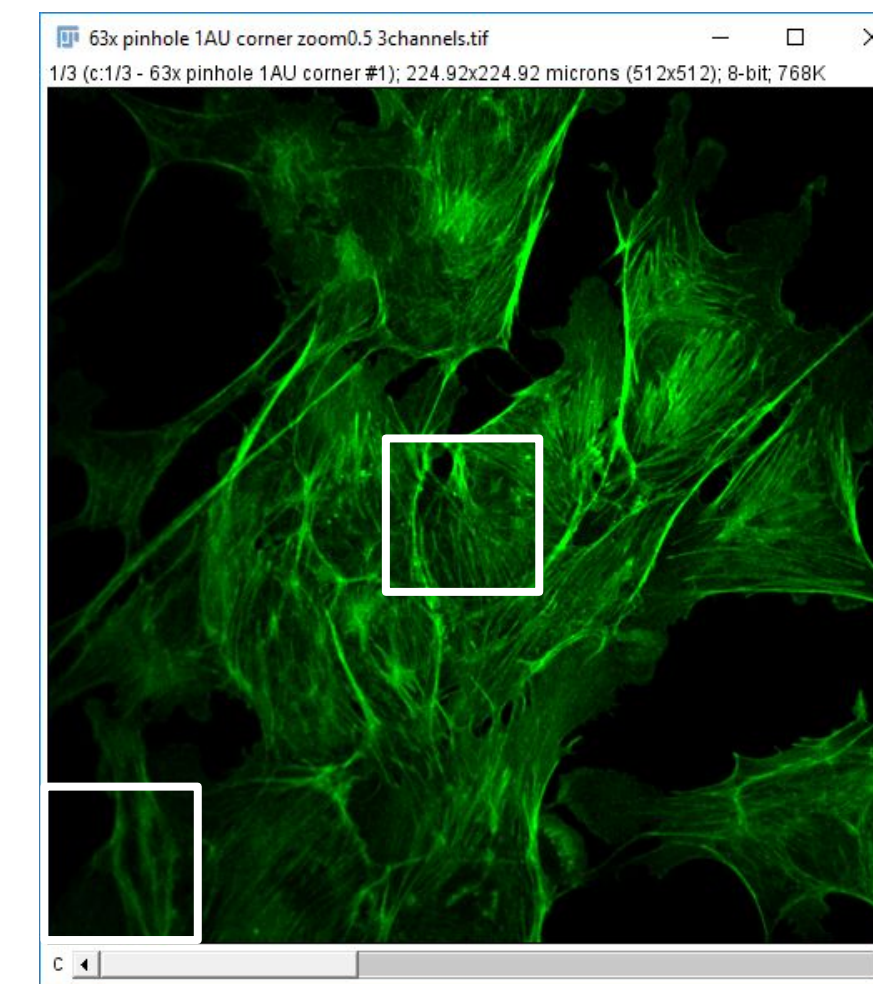
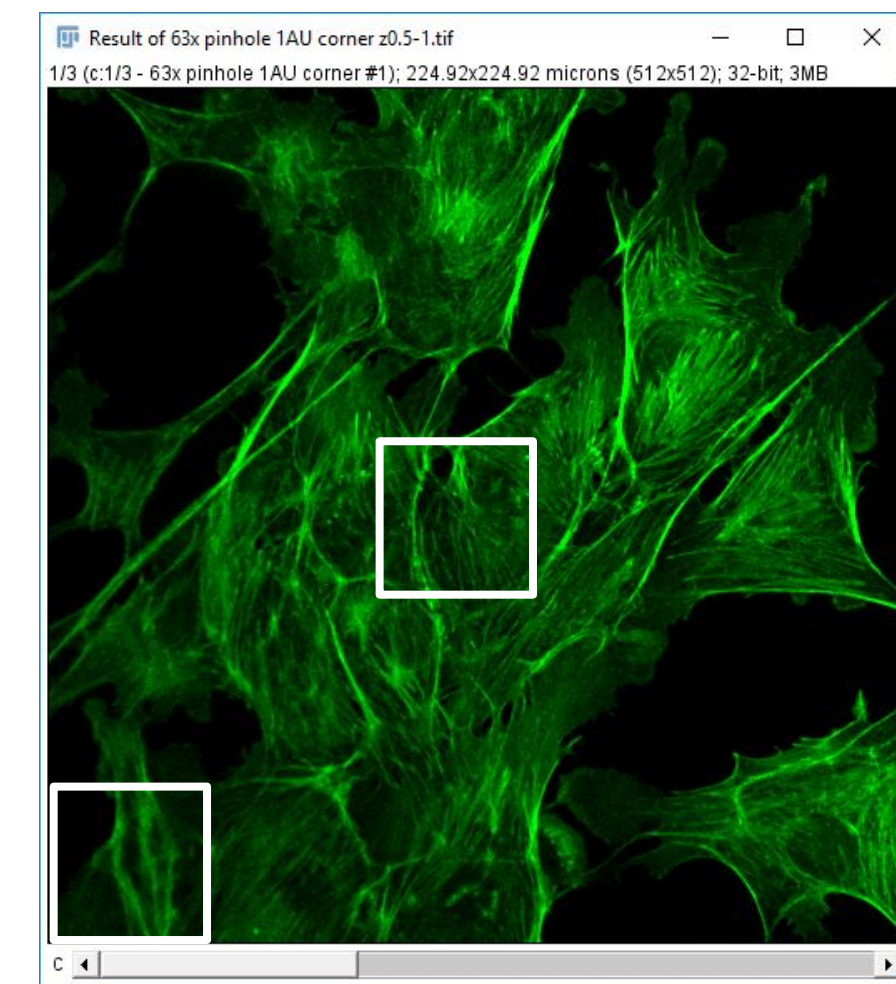


Image to Image Operations

1. Notice the gain in intensities in the corners. Note however, that these corners look noisier than the center of the image.
2. You will notice that on other channels (Like nuclei), the correction is not as powerful



Original

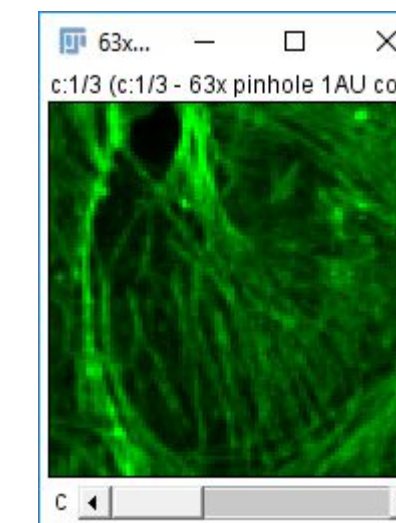


Corrected

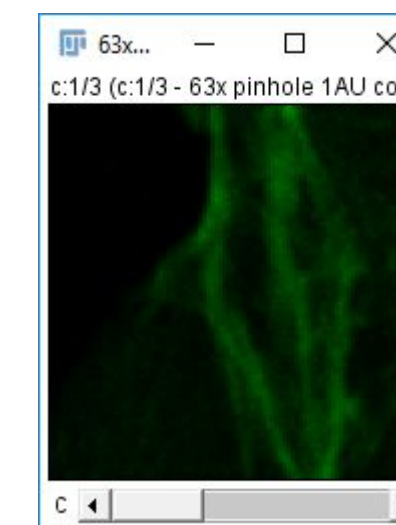
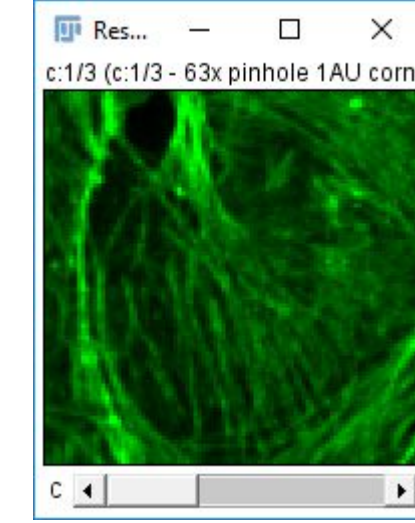
This technique can help mitigate for uneven illumination, but introduces different artifacts. You now need to deal with the fact that **your image edge does not have the same quality as your image center (More noise)**, *even though* they have recovered similar intensities.

As a rule of thumb, if edge intensities vary more than 30% from center intensities, you should crop your field of view, as subsequent analyses will most likely be biased.

See https://c4science.ch/w/bioimaging_and_optics_platform_biop/teaching/field_illumination/ for more information.



Center



Edge

